PCI AI · FORMULA SHEET

Two Languages, One Project

The complete formula reference for the PCI AI Project Controls Leader — IFRS and IAS on one side, earned value and critical path on the other

CPI · SPI · EAC · TCPI · ES · TF · EMV · CCC

PROJECT CONTROLS INSTITUTE GLOBAL

First edition · 2026 · AI proposes. The professional disposes.

WHY THIS CREDENTIAL EXISTS

A chartered accountant cannot build a schedule

They can tell you exactly when revenue is recognised under **IFRS 15**, how a provision is measured under **IAS 37**, and which borrowing costs qualify for capitalisation under **IAS 23**.

Ask them to build a time-phased cost baseline, compute an estimate at completion, or read a critical path, and the training stops.

> **This is not a criticism.** It is the boundary of a syllabus. ACCA, ICAEW, CPA and CIMA never claimed to teach project delivery.

THE MIRROR IMAGE

A planning engineer cannot close a period

They can build a four-thousand-activity network, defend the float, compute earned value under three different measurement rules and forecast an outturn three ways.

Ask them when the revenue on that work becomes recognisable, what a contract asset is, or why the cost report will not reconcile to the accounts, and the training stops.

> **Also not a criticism.** PMI, AACE and the engineering degrees never claimed to teach financial reporting.

THE PROBLEM

And the project sits in the gap

Every capital project is measured twice. Once in the language of delivery — earned value, float, forecast. Once in the language of the accounts — recognised revenue, provisions, contract assets.

The two must reconcile to the same reality. Nobody is trained to make them.

> **This is where projects are lost.** Not in the schedule, and not in the ledger — in the space between them, where each profession assumes the other has it covered.

THE ANSWER

What the PCL-AI credential certifies

EXHIBIT 1

AREA	WEIGHT	WHAT IT COVERS
A · Financial reporting and accounting	**40%**	**IFRS 15, IAS 37, IAS 23, IAS 16, IAS 2, IFRS 16, IAS**
B · Project management, controls and delivery	**40%**	**Budgeting, earned value, forecasting, sche**
C · AI for project controls	**20%**	**Governed use, with the**

Thirteen domains. Sixty-one Knowledge Areas.

> **The 40/40/20 split is the entire argument.** A controls professional who cannot follow a cost from commitment through accrual to recognised revenue cannot explain their own numbers to a board — and a finance professional who cannot read a schedule cannot challenge the forecast behind the accounts.

HOW TO USE THIS REFERENCE

Every formula in the discipline

Both languages, in full, with the mistake each formula attracts. Every entry cites the Knowledge Area where the Body of Knowledge works it through.

Standards are named at principle level and described in the Institute's own words — never reproduced.

Save this. It is the complete reference, not a highlight reel.

The language of the accounts

Domains 1 and 2. What a chartered accountant knows, stated for a controls professional.

DOMAIN 1 · THE ACCOUNTING MODEL

The model everything rests on

EXHIBIT 2

FORMULA	MEANING	KA
$A = L + E$	The accounting equation	1.1.1
Σ Debits = Σ Credits	Double-entry invariant	1.1.3
Retained earnings = Opening + Income − Expenses − Distributions	Equity movement	1.1.1

A trial balance that balances proves arithmetic, not truth. Both sides can be equally and identically wrong — which is exactly what an un-booked accrual does to a cost report.

DOMAIN 1–2 · MEASUREMENT

IAS 16 and IAS 2 — assets and inventory

EXHIBIT 3

FORMULA	STANDARD	KA
Depreciation = (Cost − Residual) ÷ Useful life	**IAS 16** Property, Plant and Equipment	1.3.4
Carrying amount = Cost − Accumulated depreciation − Impairment	**IAS 16**	1.3.4
Inventory = lower of cost and net realisable value	**IAS 2** Inventories	2.4

IAS 16 decides what is capitalised and what is expensed — and that decision moves cost between the balance sheet and the profit and loss without any change to the project itself. A controls professional who does not know where the line sits cannot explain a variance that was caused by an accounting judgement.

DOMAIN 1 · IAS 37

IAS 37 — provisions and onerous contracts

EXHIBIT 4

FORMULA	MEANING	KA
Expected value = Σ (probability_i × outcome_i)	Provision over a large population	1.4.3
Present value = Future amount ÷ (1 + r)^n	Discounting a provision	1.4.3

Two traps. Expected value applies to a **population**; for a single obligation the most likely amount is the measure, and expected value returns a figure that will never be paid. And when a contract becomes onerous, IAS 37 requires the loss to be recognised **in full, immediately** — not spread across the remaining programme. That is the accounting mirror of a forecast overrun.

DOMAIN 2 · IAS 23

IAS 23 — borrowing costs

Capitalised borrowing cost = Weighted-average qualifying expenditure × rate

Interest on funds borrowed to build a qualifying asset is capitalised into the asset, not expensed.

Capitalisation must be suspended during extended periods in which active development is interrupted. A project sitting idle still accrues interest — but that interest stops being part of the asset, and starts hitting the profit and loss. Very few controls functions track this.

DOMAIN 2 · THE FLAGSHIP KNOWLEDGE AREA

IFRS 15 — the five-step model

EXHIBIT 5

	THE FIVE STEPS
1	Identify the contract
2	Identify the performance obligations
3	Determine the transaction price
4	Allocate the price to the obligations
5	Recognise revenue as, or when, each obligation is satisfied

IFRS 15 replaced **IAS 11** Construction Contracts. The percentage-of-completion habit survived; the standard that authorised it did not.

> **This is the single largest Knowledge Area in the credential** — around 12% of the examination. It is also the one most project professionals have never been taught.

DOMAIN 2 · REVENUE

IFRS 15 — the arithmetic

EXHIBIT 6

FORMULA	MEANING	KA
PoC = Costs incurred ÷ Total estimated costs	Cost-to-cost input method	2.2.6
Cumulative revenue = PoC × Transaction price	Over-time recognition	2.2.6
Allocated price_i = Transaction price × (SSP_i ÷ Σ SSP)	Allocation by standalone selling price	2.2.5
Period revenue = Cumulative revenue − Previously recognised	What actually books this month	2.2.6

PoC is cumulative and so is the revenue it produces. Forget the fourth line and you recognise the same revenue twice. And uninstalled materials sitting in the numerator inflate progress that has not physically happened — which is why IFRS 15 requires them to be excluded from the measure.

DOMAIN 7 · CONTRACT ASSETS AND LIABILITIES

The position the contract creates

Contract asset / (liability) = Revenue recognised − Amounts billed

Recognise more than you have billed and you carry a **contract asset**. Bill more than you have earned and you carry a **contract liability**.

This single line is where the two languages meet. The left-hand term comes from IFRS 15. The right-hand term comes from the valuation your commercial team submitted. If your controls function cannot compute it, the finance team is deriving your project's position without you.

DOMAIN 2 · COMPLETING THE PICTURE

IFRS 16 and IAS 1

EXHIBIT 7

STANDARD	WHAT IT GOVERNS	KA
IFRS 16 Leases	Right-of-use asset and lease liability — plant and accommodation move onto the balance sheet	2.4
IAS 1 Presentation	The structure of the statements a project's numbers land in	1.2, 2.1
IAS 8	Accounting policies, changes in estimate and errors	2.4

IFRS 16 changed project cost structures without changing a single project. Operating leases that once sat in the cost line became assets and liabilities — which moved the same site cabin from opex to depreciation plus interest.

II

The language of delivery

Domains 3 to 12. What a planning engineer knows, stated for a finance professional.

DOMAIN 3 · BUDGET

The baseline

EXHIBIT 8

FORMULA	MEANING	KA
BAC = Σ control-account budgets + contingency	Cost baseline	3.1.4
Total authorised budget = BAC + Management reserve	Full authorisation	3.1.4
Analogous estimate = Past cost × (this driver ÷ past driver)	Analogous estimating	3.2.2
Parametric estimate = Parameter × Rate	Parametric estimating	3.2.2

Management reserve sits outside BAC. Fold it in and the baseline stops measuring anything, because you have quietly granted yourself a contingency nobody approved. AACE estimate classes govern how accurate any of this is entitled to be.

DOMAIN 5 · COST CONTROL

The three states of a cost

EXHIBIT 9

FORMULA	MEANING	KA
Total cost = Fixed + (Variable per unit × Volume)	Cost behaviour	5.1.1
OAR = Budgeted overhead ÷ Budgeted activity base	Overhead absorption rate	5.1.3
Over/(under) absorption = Absorbed − Incurred	Absorption variance	5.1.3
Cost to date = Actuals + Accruals	True cost to date	5.2.1

Commitment, then accrual, then actual. Only the third has an invoice. If **AC** carries invoiced cost alone, every index below is wrong in the same direction — and the accrual you failed to book is the same number the accountant needs for cut-off.

DOMAIN 4 · VARIANCE ANALYSIS

The variance bridge

EXHIBIT 10

FORMULA	MEANING	KA
Price variance = (Actual price − Standard price) × Actual quantity	Price effect	4.2.3
Quantity variance = (Actual qty − Standard qty) × Standard price	Quantity effect	4.2.3
Total variance = Price variance + Quantity variance	Reconciliation check	4.2.3

Price takes actual quantity; quantity takes standard price. Swap the multipliers and the two will not sum to the total — which is precisely why the third line is on the sheet.

DOMAIN 6 · THE INDICES

Earned value — position

EXHIBIT 11

FORMULA	MEANING	KA
CV = EV − AC	Cost variance	6.2.1
SV = EV − PV	Schedule variance, in currency	6.2.1
CPI = EV ÷ AC	Cost efficiency to date	6.2.2
SPI = EV ÷ PV	Schedule efficiency in cost terms	6.2.2
% complete = EV ÷ BAC	Progress by value	6.1

SV and SPI are denominated in money, not time. At completion EV = PV, so both return to zero and 1.00 however late you finished. Honest early, misleading late.

DOMAIN 6 · THE EAC FAMILY

Earned value — forecast

EXHIBIT 12

FORMULA	THE ASSUMPTION IT ENCODES	KA
EAC = AC + ETC	Identity — always true	6.3.1
EAC = AC + (BAC − EV)	The variance was a one-off	6.3.2
EAC = BAC ÷ CPI	Performance to date persists	6.3.2
EAC = AC + (BAC − EV) ÷ (CPI × SPI)	Cost and schedule both persist	6.3.2
VAC = BAC − EAC	Variance at completion	6.3.4
TCPI = (BAC − EV) ÷ (BAC − AC)	Efficiency required to hit budget	6.2.3

The arithmetic is trivial; the assumption is the judgement. Set **TCPI** beside the **CPI** you have achieved — a **TCPI** far above it asserts a gain nobody has demonstrated. And an identity worth knowing: when **EAC = BAC ÷ CPI**, **TCPI (to EAC)** equals **CPI** exactly.

DOMAIN 6 · EARNED SCHEDULE

Schedule performance in time

EXHIBIT 13

FORMULA	MEANING	KA
ES = M + (EV − PV_M) ÷ (PV_M+1 − PV_M)	Interpolate between the cumulative-PV periods bracketing **EV**	6.4.3
SV(t) = ES − AT	Schedule variance in time	6.4.3
SPI(t) = ES ÷ AT	Time-based schedule efficiency	6.4.3

These do not converge at completion, which is what makes them the honest late-stage measure. Use cumulative planned value in the interpolation, never the period figure.

DOMAIN 10 · CRITICAL PATH

The network

EXHIBIT 14

FORMULA	MEANING	KA
EF = ES + Duration	Forward pass	10.2
LS = LF − Duration	Backward pass	10.2
Total float = LS − ES = LF − EF	Slack without delaying the project	10.2.4
Free float = min(successor ES) − EF	Slack without delaying a successor	10.2.4
tE = (O + 4M + P) ÷ 6 · σ = (P − O) ÷ 6	PERT three-point estimate	10.1.4
σ^2 path = Σ σ^2 along the path	Path variance	10.3.4

Total float from the starts must equal total float from the finishes. If they disagree, a pass is wrong. And variances add along a path — standard deviations do not.

DOMAIN 12 · ISO 31000

Risk and contingency

EXHIBIT 15

FORMULA	MEANING	KA
EMV = Probability × Impact	Expected monetary value	12.2.3
Total EMV = Σ (P_i × I_i)	Portfolio exposure	12.2.3
Contingency ≈ Σ EMV, or P80 − P50	Risk-based contingency	12.3.1

A percentage of budget is a habit, not an analysis. And note where this lands in the accounts: a risk that becomes probable and measurable stops being contingency and becomes an **IAS 37 provision**. The two professions are describing the same event with different words.

DOMAIN 9 · AGILE

Adaptive delivery

EXHIBIT 16

FORMULA	MEANING	KA
% complete = Points completed ÷ Total planned points	AgileEVM progress	9.5.3
EV = % complete × BAC	Earned value from points	9.5.3
Velocity = Points completed ÷ Sprint	Delivery rate	9.3
Sprints remaining = Points remaining ÷ Velocity	Forecast to completion	9.3
Run rate = Team cost ÷ Sprint	Cost per sprint	9.5.2
Cycle time = WIP ÷ Throughput	Little's Law	9.4

Point inflation breaks the first two lines, and it is invisible unless someone is watching. Note that IFRS 15 still applies to agile delivery — the standard does not care how you organised the work.

DOMAINS 7, 11 · CONTRACT AND WORKING CAPITAL

Commercial and cash

EXHIBIT 17

FORMULA	MEANING	KA
Fee = Target fee + Contractor share × (Target − Actual cost)	CPIF incentive fee	7.1.3
Pain/gain = Share ratio × (Actual − Target)	Target-cost mechanism	7.1.4
LD exposure = LD rate × Days late	Liquidated damages	7.2.3
Amount due = Σ(% complete × item) − Retention − Previous payments	Interim application	7.4.3
DSO = Receivables ÷ Daily revenue · DPO = Payables ÷ Daily COGS	Working capital days	11.1.3
CCC = DSO + DIO − DPO	Cash conversion cycle	11.A.1

Liquidated damages are the commercial event; IAS 37 is the accounting consequence. Once they are probable and measurable, they are a provision — recognised now, not when the deduction lands.

Where the two languages meet

The integration. This is what neither profession is taught, and what the credential exists for.

THE INTEGRATION SPINE

One cost, two destinations

A cost is incurred on site. It then travels two paths at once.

EXHIBIT 18

PATH	SEQUENCE
Delivery	Commitment → accrual → **AC** → **CPI** → **EAC**
Accounts	Accrual → **PoC** → cumulative revenue → contract asset or liability

The accrual is the shared node. Both paths depend on it, and only one profession is usually watching it.

> **Fail to book it and two things happen simultaneously:** **CPI** overstates delivery performance, and **PoC** understates revenue. The project looks efficient and under-billed at the same time — a combination that reads as good news in both languages.

THE TEST

The reconciliation nobody owns

EXHIBIT 19

QUESTION	LANGUAGE	WHO ANSWERS IT
What has this cost us?	Delivery	**Project controls**
What may we recognise?	Accounts	**Finance**
Do those two rest on the same accrual?	**Both**	**Nobody, usually**

A standing unreconciled difference between the cost report and the ledger is not a timing quirk. It is two sets of books, and it will be discovered by an auditor rather than by you.

A TRANSLATION TABLE

The same event, two vocabularies

EXHIBIT 20

THE DELIVERY LANGUAGE	THE ACCOUNTING LANGUAGE
Contingency drawdown for a probable risk	**IAS 37** provision
Forecast overrun on a fixed-price contract	**IAS 37** onerous contract — recognise the loss in full, now
Progress claimed but not yet certified	**IFRS 15** contract asset
Advance payment received	**IFRS 15** contract liability
Interest on funds during the build	**IAS 23** capitalised borrowing cost
Site accommodation lease	**IFRS 16** right-of-use asset
Cost to date including accruals	The cut-off the accounts depend on

Every row is one event described twice. A professional fluent in only one column is guessing at the other — and boards ask questions in both.

WHERE A FORECAST BECOMES A LOSS

The onerous contract test

When **EAC** exceeds the contract value, the delivery language calls it a negative **VAC**.

The accounting language calls it an **onerous contract** under IAS 37 — and requires the entire expected loss to be recognised **immediately**, not spread over the remaining programme.

> **This is the highest-stakes translation in the discipline.** A controls professional who reports a deteriorating EAC without knowing it triggers an immediate provision has told the board half of the news. The half they left out is the half that hits this period's results.

IV

Reference

AT PRINCIPLE LEVEL, NEVER REPRODUCED

Standards named in this credential

EXHIBIT 21

STANDARD	GOVERNS	DOMAINS
IFRS 15	Revenue from contracts with customers	2.2, 2.3, 7.5, 9.5
IAS 37	Provisions, contingent liabilities, onerous contracts	1.4, 2.2, 2.4
IAS 16	Property, plant and equipment	1.3, 2.4
IAS 23	Borrowing costs	2.4
IAS 2	Inventories	2.4
IFRS 16	Leases	2.4
IAS 1	Presentation of financial statements	1.2, 2.1
IAS 11 (legacy)	Construction contracts — superseded by IFRS 15	2.4
ISO 31000	Risk management	12
AACE TCM	Total cost management, estimate classes	3.2
PMBOK Guide	Process groups and practices	8

MOST MISAPPLIED · 1–5

The ten that cost the most

1. **CPI** computed on invoiced cost, with no accrual booked
2. **SV** read as a schedule measure late in a project
3. A single **EAC** presented as the forecast
4. A deteriorating **EAC** reported without the **IAS 37** onerous-contract consequence
5. Percentage complete supplied by the party being measured

MOST MISAPPLIED · 6–10

And the other five

1. **PoC** computed with uninstalled materials in the numerator
2. Cumulative revenue booked without deducting what was already recognised
3. Contingency set as a percentage instead of quantified exposure
4. Standard deviations summed along a schedule path
5. Management reserve folded inside **BAC**

> **Note how the list divides.** Five are delivery errors, five are accounting errors, and every one of them lands on the same project. That division is the argument for this credential in a single slide.

Learn both. Nobody else teaches both.

Accountancy teaches the accounts. Engineering teaches delivery. The **PCI AI Project Controls Leader** teaches the two together, because a capital project is measured in both and reconciled by whoever can speak them.

Thirteen domains, sixty-one Knowledge Areas, weighted 40% finance, 40% delivery, 20% governed AI. The Body of Knowledge is published openly — **no registration, no email gate**.

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